

Matrices

① Row and Column Matrix

Row Matrix मतलब बस EK

Row Hai $[1 \ 2 \ 3]$

Column Matrix मतलब बस EK Column Hai

④ Trace of a Matrix

Trace of any = Sum of its diagonal elements
Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A \quad \text{Trace}(A) = 3$$

⑤ Singular Matrix

aisi matrix jiske det ka Value 0 Ho

⑨ Scalar Matrix

aisi diagonal Matrix jisme diagonal elements same Ho

$$\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

⑪ Triangular Matrix

① Lower Δ Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 3 \end{bmatrix}$$

Jisme row में Δ बने और upper 0

⑬ Symmetric Matrix

• aisi matrix jaha diagonal element non zero Ho aur they act as mirror

• Mirror ke left aur right same element

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & -6 \\ 5 & -6 & 3 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & -6 \\ 5 & -6 & 3 \end{bmatrix} \quad A = A^T$$

property of Symmetric matrix

$$A = A^T$$

② Square matrix

no of rows = no of column
n x n order

$$\begin{bmatrix} 2 & 9 \\ 0 & 2 \end{bmatrix}_{2 \times 2}$$

⑥ Non singular Matrix

aisi matrix jiske det ka value non zero Ho

⑦ Unit / Identity Matrix

$$I_{2 \times 2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_{3 \times 3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

⑩ Transpose of a Matrix

To get Transpose of a Matrix, say 'A' swap the rows and column
Row ko column aur Column ko row
 $A \rightarrow A^T$

③ Diagonal Matrix

Aisi matrix jisme Non diagonal elements 0 Hai aur diagonal elements non 0

det $[a_{ij}]_{m \times n}$ be a matrix
If $[a_{ij}]_{m \times n} = \begin{cases} a_{ij} = 0, & i \neq j \\ a_{ij} \neq 0, & i = j \end{cases}$

then $[a_{ij}]_{m \times n}$ is said to be diagonal matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

⑧ Zero/Null Matrix

all elements are zero

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$a_{ij} = 0 \quad \forall i, j$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 9 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 9 \\ 3 & 1 \end{bmatrix}$$

⑫ Conjugate Matrix

Used for Matrices Containing complex numbers

$$A = \begin{bmatrix} 2 & 2+i \\ 3i & 4+5i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 2 & 2-i \\ -3i & 4-5i \end{bmatrix}$$

A ka Conjugate matrix ke line aur ke Complex No ka Conjugate lelo

⑭ Skew Symmetric Matrix

• aisi matrix jisme diagonal elements 0 Ho aur they act as $\ominus$ Mirror
• Mirror ke left right opposite sign ke element

$$A = \begin{bmatrix} 0 & -4 & -1 \\ 4 & 0 & 2 \\ 1 & -2 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 0 & 4 & 1 \\ -4 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix}$$

$$\text{Here } A^T = -A$$

⑮ Hermitian Matrix

Matrix whose Transpose conjugate matrix equal to it

$$A^H = A, \quad (\bar{A})^T = A$$

If $(\bar{A})^T \neq A$, Then Skew Hermitian Matrix



Orthogonal Matrix

If a matrix is orthogonal if $A \cdot A^T = I$

If a matrix is orthogonal then $A^{-1} = A^T$

Que) $A = \frac{1}{3} \begin{bmatrix} -2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{bmatrix}$

$A^T = \frac{1}{3} \begin{bmatrix} -2 & 2 & 1 \\ 1 & 2 & -2 \\ 2 & 1 & 2 \end{bmatrix}$

$A \cdot A^T = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

$\Rightarrow A \cdot A^T = I$

$\therefore A$ is a orthogonal matrix

$\therefore A^{-1} = A^T$

que) If ψ is orthogonal find a, b, c

$\psi = \frac{1}{3} \begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix}$

$\psi^T = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & b & c \end{bmatrix}$

$\psi \cdot \psi^T = I \Rightarrow \frac{1}{9} \begin{bmatrix} a^2+5 & 4+ab & ac \\ 4+ab & b^2+5 & 2+bc \\ ac-2 & bc+2 & c^2+8 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

$a^2+5=9$

$a^2=4$

$a=\pm 2$

$b^2+5=9$

$b^2=4$

$b=\pm 2$

$c^2+8=9$

$c^2=1$

$c=\pm 1$

que) $A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & a \\ \frac{2}{3} & \frac{1}{3} & b \\ \frac{2}{3} & -\frac{2}{3} & c \end{bmatrix}$

$A^T = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ a & b & c \end{bmatrix}$

$AA^T = \begin{bmatrix} \frac{1}{9} + \frac{4}{9} + a^2 & \frac{4}{9} + \frac{1}{9} + b^2 & \frac{4}{9} + \frac{4}{9} + c^2 \\ \frac{4}{9} + \frac{1}{9} + b^2 & \frac{4}{9} + \frac{4}{9} + c^2 & \frac{4}{9} + \frac{1}{9} + b^2 \\ \frac{4}{9} + \frac{4}{9} + c^2 & \frac{4}{9} + \frac{1}{9} + b^2 & \frac{4}{9} + \frac{4}{9} + c^2 \end{bmatrix}$

$5+9a^2=9$

$9a^2=4$

$a^2=\frac{4}{9}$

$a=\pm \frac{2}{3}$

$5+9b^2=9$

$b^2=\frac{4}{9}$

$b=\pm \frac{2}{3}$

$8+9c^2=9$

$9c^2=1$

$c=\pm \frac{1}{3}$

① Unitary Matrix

agar kisi Matrix aisi hai like $A \cdot A^H = I$, then A is said to be Unitary matrix.

If $A \cdot A^H = I$ then A is said to be Unitary Matrix

$$A^H = (\bar{A})^T$$

Que) PT $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary

$$A^T = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1-i \\ 1+i & -1 \end{bmatrix} \quad (A^T)^C = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$A \cdot A^T = \frac{1}{3} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1+1+1 & 0 \\ 0 & 1+1+1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$\therefore A$ is a unitary Matrix

Que) PT $A = \begin{bmatrix} \frac{1+i}{2} & -\frac{1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ is a unitary matrix

$$A = \frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix} \quad A^T = \begin{bmatrix} 1+i & 1+i \\ -1+i & 1-i \end{bmatrix}$$

$$(\bar{A}^T) = \frac{1}{4} \begin{bmatrix} 1-i & 1-i \\ -1-i & 1+i \end{bmatrix} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2+2 & 2+2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} & - (1-i)^2 \\ & 1-i \\ & (1-i)(i-1) \\ & - (1-i)^2 \\ & (1+i)(1-i) - (i+1) \end{aligned}$$

Rank of a Matrix

Rank of Matrix पता करने के लिये final Matrix की Rows देखी → अगर Rows में 0 के alawa कोई और No say! hai → Toh No of Rows containing a non zero element is called Rank of Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Rank} = 2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Rank} = 3$$

जिस Row में data hai
Voh Row गिन ली।

1) We can swap any two rows or columns
 $R_2 \leftrightarrow R_1$ (usually done for making 1)

2) Elementary Row/ column transformations

$$R_1 \rightarrow R_1 + R_2 \text{ etc}$$

3) Row/ Column ko किसी scalar se divide

$$R_4 \rightarrow R_4 / 20$$

• Method for Ranks

① Normal Form

② Echelon Form

□ Normal Form

① $\begin{bmatrix} 1 & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$ Step 1 → left hand पर '1' बनाओ
"हमेशा" *swapping/divide

② $\begin{bmatrix} 1 & \rightarrow 2 \\ & 1 \\ & 0 \\ & 0 \end{bmatrix}$ Step 2 → 1 के नीचे और straight देखने पर नो दिखे उन सबको '0' बना दो।

③ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ Step 3
fir diagonally नीचे उतरते और वही 1 बनाओ, fir again नीचे और straight की 0 करो

→ तब तक जब तक Identity Matrix जैसा ना हो जाये

□ Echelon Form

* only Row transformation will be used

$\begin{bmatrix} 1 & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$ → यहाँ पर 1 लेकर आओ और उसके नीचे के सब को 0 करो

$\begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & 0 & 0 \end{bmatrix}$ → अब यहाँ 1 बनाओ और इसके नीचे 0 बनाओ!

इसमें बस नीचे देख कर 0 बनाओ, सिधे सुझ कर मत देखना।

* यहाँ हमेशा maximum 0 वाली rows को • में Bottom में arrange करेंगा।

③ Reduction of Matrix to normal form

System of non Homogeneous equations

$\begin{bmatrix} 1 & 3 & 2 \\ -4 & 2 & 5 \\ 3 & 4 & 6 \end{bmatrix}$
 → यहाँ ये 1 बनाना जरूरी नहीं है।
 * don't apply Column transformation
 → इन सबको 0 बना दो
 * only apply Row Transformation

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 5 & 9 & 1 \\ 6 & 0 & 2 \end{bmatrix}$$

0 बनाने की Ninja technique when 1 is Not there : जो तेरा है वो मेरा है

$$\begin{bmatrix} 1 & 3 & 3 \\ 0 & 2 & 9 \\ 0 & 7 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 - 7R_2$$

Non Homogeneous eqⁿ

If Rank of A $\neq$ Rank of AB
 Then Solution is inconsistent
 $\Rightarrow$ No Solution

$$AX = B$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

$\sigma = \text{rank of } A$

Rank of AB = combine AB
= 3

$n = \text{no of variable (xyz)}$

$$\begin{bmatrix} 1 & 2 & 3 & : & 12 & 3 \\ 0 & 1 & 2 & : & 0 & 2 & 1 \\ 0 & 0 & 0 & : & 2 & 0 & 1 \end{bmatrix}$$

If Rank of A = Rank of AB
 Then Solution of Consistent

If $\sigma = n$
 Unique solution
 Trivial solution

If $\sigma < n$
 Infinite solution
 Trivial solution

Que) Test the consistency and solve it

$$x + y + z = 6$$

$$x + 2z - y = 5$$

$$3x + y + z = 8$$

$$2x - 2y + 3z = 7$$

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \\ 2 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \\ 8 \\ 7 \end{bmatrix}$$

How to Convert to

$$AX = B ?$$

A = Coeff of x, y, z in order
की matrix

B = RHS ke Constant

* AB exist If *

$$A = m \times n$$

$$B = n \times p$$

ye match hua kaly

multiply hake order = $m \times p$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 1 \\ 2 & -2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & -2 & -2 \\ 0 & -4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & -1 \end{bmatrix}$$

$$R_4 \rightarrow 3R_4 + R_3$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} X = \begin{bmatrix} 6 \\ -1 \\ -9 \\ 0 \end{bmatrix}$$

Rank of A = Rank of AB
and $\sigma = n$

ये सब operation d_1, d_2, d_3, d_4 $\therefore$ Trivial solutions
value matrix पर भी लागेगा।



$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \\ -9 \\ 0 \end{bmatrix}$$

4x3 3x1

4 Row 1 column

$$\begin{bmatrix} x+y+z \\ -2y+z \\ -3z \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \\ -9 \\ 0 \end{bmatrix}$$

$$\begin{aligned} z &= 3 \\ -2y+3 &= -1 \\ -2y &= -4 \\ y &= 2 \end{aligned}$$

$$\begin{aligned} -2 + x + x &= 6 \\ x &= 5 \end{aligned}$$

(5, -2, -1)

Que) Test for consistency and solve if consistent

$$x - 2y + 3z = 2$$

$$2x + y + z + t = 4$$

$$4x - 3y + z + 7t = 8$$

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 2 & 1 & 1 & 1 \\ 4 & -3 & 1 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 0 & 5 & 1 & -5 \\ 0 & 5 & 1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & -2 & 0 & 3 \\ 0 & 5 & 1 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Rank of } A = 2 \quad \text{Rank of } AB = 2$$

$n < n$ ∴ Infinite soln

Q) for what value of λ eqⁿ have solution, find solⁿ too

$$x + y + z = 1$$

$$x + 2y + 4z = \lambda$$

$$x + 4y + 10z = \lambda^2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \lambda \\ \lambda^2 \end{bmatrix}$$

have solⁿ
= rank = n

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \lambda - 1 \\ \lambda^2 - 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ \lambda - 1 \\ \lambda^2 - 2\lambda + 1 \end{bmatrix}$$

$$\begin{aligned} \lambda^2 - 1 - 2(\lambda - 1) \\ \lambda^2 - 1 - 2\lambda + 2 \\ \lambda^2 - 2\lambda + 1 \end{aligned}$$

$$\text{rank of } A = 2 \quad \text{rank of } AB = 2$$

$$\therefore \lambda^2 - 2\lambda + 1 = 0$$

$$(\lambda - 1)^2 = 0 \Rightarrow \lambda = 1$$

$$\lambda^2 - 3\lambda + 2$$

$$(\lambda - 2)(\lambda - 1) = 0$$

$$\lambda = 2$$

for $\lambda = 1$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

3x3 3x1
3 row 1c

$$\begin{bmatrix} x + y + z \\ y + 3z \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x + y + z = 0$$

$$y = -3z$$

$$x - 3z + z = 0$$

$$x = 2z$$

for $\lambda = 2$

$$\begin{bmatrix} x + y + z \\ y + 3z \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$y + 3z = 1$$

$$r < n$$

Q) Type 3: Danger question

for what value of λ and μ eqⁿ have no solⁿ, unique solⁿ, ∞ solⁿ

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & \lambda - 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ \mu - 10 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & \lambda - 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ \mu - 10 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$



Homogeneous equation

Homogeneous equations $\rightarrow$ Take $d_1, d_2, d_3 = 0$ eq 11

no of eqⁿ $\neq$ no of variable

no of eqⁿ = no of variable

$A = \text{Coeff Matrix}$

$\text{rank}(A) = \text{no of var}$

- Unique solⁿ
- Trivial solⁿ

$\text{rank}(A) < \text{no of var}$

- ∞ solutions
- Non Trivial solution

$\det(A) = 0$

- ∞ solution
- Non Trivial solⁿ

$\det(A) \neq 0$

- Unique solution
- Solⁿ $\rightarrow x=y=z=0$

Que) for what value of λ , the eqⁿ possesses a non trivial solution

$$3x_1 + x_2 - \lambda x_3 = 0$$

$$4x_1 - 2x_2 - 3x_3 = 0$$

$$2\lambda x_1 + 4x_2 + \lambda x_3 = 0$$

$$n(\text{eq}^n) = n(\text{var})$$

$$\det(A) = 0$$

$$\begin{vmatrix} 3 & 1 & -\lambda \\ 4 & -2 & -3 \\ 2\lambda & 4 & \lambda \end{vmatrix} = 0$$

$$3(-2\lambda + 12) - 1(4\lambda + 6\lambda) - \lambda(16 + 4\lambda) = 0$$

$$3(12 - 2\lambda) - 10\lambda - 16\lambda - 4\lambda^2 = 0$$

$$-4\lambda^2 - 26\lambda - 6\lambda + 36 = 0$$

$$4\lambda^2 + 32\lambda + 36 = 0$$

$$\lambda^2 + 8\lambda + 9 = 0$$

$$\lambda^2 + 9\lambda - \lambda + 9 = 0$$

$$\lambda(\lambda + 9) - (\lambda + 9) = 0$$

$$(\lambda - 1)(\lambda + 9) = 0$$

$$\lambda = -9$$

$$\lambda = 1$$

Que) solve

$$x_1 + x_2 - x_3 + x_4 = 0$$

$$x_1 - x_2 + 2x_3 - x_4 = 0$$

$$3x_1 + x_2 + x_4 = 0$$

$$n(\text{var}) > n(\text{eq}^n)$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 + C_1$$

$$C_4 \rightarrow C_4 - C_1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 3 & -2 \\ 0 & -2 & 3 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$C_4 \rightarrow C_4 - C_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -2 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{rank}(A) = 2 < \text{no of eq}^n(\text{var})$$

∞ solⁿ



③ Inverse of Matrix By Gauss Jordan Method

agar 'A' matrix का Inverse निकालना है

get AI_n where $n = \text{order of } A$

Now AI में A me Row/Column Transformation लाके use I

बनादी (operation I में भी लगेंगी)

अब जब tumhe A की I बना दिया है, I जो है वो A^{-1} बन चुका है।

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 2 & -6 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 + 2R_1$$

$$-4+6$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 - 3C_1$$

$$1-3(2)$$

$$1-6$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & -6 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & -3 \\ -1 & 2 & 3 \\ 2 & -2 & -5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 & -3 \\ -1/2 & 1 & 3/2 \\ 2 & -2 & -5 \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{2}$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$2 + 2(-\frac{1}{2})$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 & -1 & -3 \\ -1/2 & 1 & 3/2 \\ 1 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 & -1 & -6 \\ -1/2 & 1 & 9/2 \\ 1 & 0 & -2 \end{bmatrix}$$

$$C_3 \rightarrow C_3 + 3C_2$$

$$R_3 \rightarrow \frac{R_3}{(-4)}$$

$$2-1$$

$$-2+2(1)$$

$$-5+2(\frac{3}{2})$$

$$-5+3=-2$$

$$-3+3(-1)$$

$$\frac{3}{2} + 3(1)$$

$$2+2(-3)$$

$$2-6$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -6 \\ -1/2 & 1 & 9/2 \\ -1/4 & 0 & 1/2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & -1 & -6 \\ -1/2 & 1 & 9/2 \\ -1/4 & 0 & 1/2 \end{bmatrix} \quad (\otimes)$$

$$A^{-1} = \begin{bmatrix} 3 & 1 & 3/2 \\ -5/4 & -1/4 & -3/4 \\ -1/4 & -1/4 & -1/4 \end{bmatrix} \quad (\checkmark)$$

⊙ Eigen Values and Eigen vectors

To get eigen value of a Matrix A

find $\det(A - \lambda I) = 0$

yaha se λ ek eqⁿ Banegi $\rightarrow$ Jiske roots will be eigen values of A

for eigen vector $[A - \lambda I][x] = 0$

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 6 & 9 & 1 \\ 5 & 4 & 3 \end{bmatrix}$$

Now $\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 3 & 5 \\ 6 & 9-\lambda & 1 \\ 5 & 4 & 3-\lambda \end{vmatrix} = 0$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

Either solve the determinant Manually By hand (Expand or ft)

★ alternate +, -

★ $S_1 = \text{trace of } A$

★ $S_2 = \text{Sum of "minor of diagonal element"}$

que) find eigen values and eigen vectors of following matrix

$$A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 8-\lambda & -8 & -2 \\ 4 & -3-\lambda & -2 \\ 3 & -4 & 1-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda_1 = 1 \quad \lambda_2 = 2 \quad \lambda_3 = 3$$

eigen value of matrix = 1, 2, 3

$$[A - \lambda I] = \begin{bmatrix} 8-\lambda & -8 & -2 \\ 4 & -3-\lambda & -2 \\ 3 & -4 & 1-\lambda \end{bmatrix}$$

for $\lambda = 1$

$$\begin{bmatrix} 7 & -8 & -2 \\ 4 & -4 & -2 \\ 3 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 7x_1 - 8x_2 - 2x_3 \\ 4x_1 - 4x_2 - 2x_3 \\ 3x_1 - 4x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Altier: Using Cramers rule

$$7x_1 - 8x_2 - 2x_3 = 0$$

$$4x_1 - 4x_2 - 2x_3 = 0$$

$$\frac{x_1}{\begin{vmatrix} -8 & -2 \\ -4 & -2 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 7 & -2 \\ 4 & -2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 7 & -8 \\ 4 & -4 \end{vmatrix}}$$

$$\frac{x_1}{4} = \frac{x_2}{3} = \frac{x_3}{2}$$

for $\lambda = 1$
eigen vector = $\begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix}$

जब x_3

लिखना तो

x_3 के coeff

का determinant

में लिख

$$\begin{matrix} 3 \times 3 & 3 \times 1 \\ 3 \times 1 \end{matrix}$$

$$7x_1 - 8x_2 - 2x_3 = 0 \dots \textcircled{2}$$

$$3x_1 = 4x_2 \dots \textcircled{1}$$

$$4x_1 - 4x_2 - 2x_3 = 0$$

$$4x_1 - 3x_1 - 2x_3 = 0$$

$$x_1 - 2x_3 = 0$$

x eqⁿ ① with 2

$6x_1 = 8x_2 \rightarrow$ put in eqⁿ ②

$$7x_1 - 6x_1 - 2x_3 = 0$$

$$\boxed{\begin{matrix} x_1 = 2x_3 \\ 6x_3 = 4x_2 \end{matrix}}$$

for $\lambda = 3$

$$[A - \lambda I] = \begin{bmatrix} 5 & -8 & -2 \\ 4 & -6 & -2 \\ 3 & -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x_1 - 8x_2 - 2x_3 = 0$$

$$4x_1 - 6x_2 - 2x_3 = 0$$

$$\frac{x_1}{\begin{vmatrix} -8 & -2 \\ -6 & -2 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 5 & -2 \\ 4 & -2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 5 & -8 \\ 4 & -6 \end{vmatrix}}$$

$$\frac{x_1}{4} = \frac{-x_2}{-2} = \frac{x_3}{2}$$

$$\frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{1}$$

for $\lambda = 3$ eigen vector $\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$$\begin{aligned} 16 - (12) \\ -10 - (4 \cdot -2) \\ -10 - (-8) \\ -10 + 8 \\ -30 + 32 \end{aligned}$$

for $\lambda = 2$

$$[A - \lambda I] = \begin{bmatrix} 6 & -8 & -2 \\ 4 & -5 & -2 \\ 3 & -4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$6x_1 - 8x_2 - 2x_3 = 0$$

$$4x_1 - 5x_2 - 2x_3 = 0$$

$$= \frac{x_1}{\begin{vmatrix} -8 & -2 \\ -5 & -2 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 6 & -2 \\ 4 & -2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 6 & -8 \\ 4 & -5 \end{vmatrix}}$$

$$= \frac{x_1}{6} = \frac{-x_2}{-4} = \frac{x_3}{2}$$

for $\lambda = 2$
Eigen vector = $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

$$= \frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{1}$$

Q) find eigen value/vector of A [EIGEN VALUE is repeated]

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$[A - \lambda I] = \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix}$$

characteristic eqn = $\lambda^3 - 7\lambda^2$

$$\det(A) = 2(4) - 2(1) + 1(2-3) = 8 - 2 - 1 = 5$$

$$\Sigma (6-2) + (4-1) + (6-2) = S_2$$

$$S_2 = 4 + 3 + 4 = 11$$

CE $\Rightarrow$

$$\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$$

$$\lambda = 1$$

$$1 - 7 + 11 - 5 = 12 - 12 = 0 \quad \lambda = 1 \text{ is a root}$$

$$(\lambda - 1)(\lambda^2 - 6\lambda + 5) = 0$$

$$(\lambda - 1)(\lambda - 1)(\lambda - 5) = 0$$

$$\lambda = 1, \lambda = 1, \lambda = 5$$

$$\lambda^2 - 5\lambda - \lambda + 5 = 0$$

$$\lambda(\lambda - 5) - (\lambda - 5)$$

$$\begin{array}{r} 1 \quad 1 \quad -7 \quad 11 \quad -5 \\ - \quad 1 \quad -6 \quad 5 \\ \hline 1 \quad -6 \quad 5 \quad 0 \end{array}$$



for $\lambda = 5$

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$[A - \lambda I] = \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix}$$

$$\begin{bmatrix} -3 & 2 & 1 \\ 1 & -2 & 1 \\ 1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{aligned} -3x_1 + 2x_2 + x_3 &= 0 \\ x_1 - 2x_2 + x_3 &= 0 \end{aligned}$$

$$\frac{x_1}{\begin{vmatrix} 2 & 1 \\ -2 & 1 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} -3 & 1 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -3 & 2 \\ 1 & -2 \end{vmatrix}}$$

$\frac{2+2}{-3-1} = \frac{6-2}{-4}$

$$\frac{x_1}{4} = \frac{-x_2}{-4} = \frac{x_3}{4}$$

$$x_1 = x_2 = x_3$$

for $\lambda = 5$ eigen vector $x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

for $\lambda = 1$ and $\lambda = 1$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

When eigen values repeat 30 case me Grammer ke $\frac{x_1}{0}$, $\infty = 0$ making it ∞ , is case me R_2 aur R_3 ko 0 karo aur ek eqⁿ nikalo

$$\begin{aligned} R_2 &\rightarrow R_2 - R_1 \\ R_3 &\rightarrow R_3 - R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

let $x_1 = t$ and $x_2 = s$

$$x_3 = -2s - t$$

eigen vectors $x = \begin{bmatrix} t \\ s \\ -2s - t \end{bmatrix}$ { Now Manipulate x into x_1 and x_2 }

$$x = \begin{bmatrix} 0s + t \\ s + 0t \\ -2s - t \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} s + \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} t$$

$$x_1 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Que) If $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ find eigen Value of $4A^{-1} + 3A + 2I$

A^{-1} by Gauss-Jordan

$$\begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$$

$$R_2 \rightarrow \frac{R_2}{4}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1/2 & 0 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1 & 0 \\ -1/2 & 0 \end{bmatrix}$$

$$4A^{-1} = \begin{bmatrix} 4 & 0 \\ -2 & 0 \end{bmatrix}$$

$$3A = \begin{bmatrix} 3 & 0 \\ 6 & 12 \end{bmatrix}$$

$$2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= 4A^{-1} + 3A + 2I = \begin{bmatrix} 9 & 0 \\ 4 & 14 \end{bmatrix} = N$$

finding eigen value of N

$$\det(A - \lambda I) = \begin{bmatrix} 9 - \lambda & 0 \\ 4 & 14 - \lambda \end{bmatrix} = 0$$

$$(9 - \lambda)(14 - \lambda) = 0$$

$$\lambda = 9 \quad \lambda = 14$$

for $\lambda = 9$

$$\begin{bmatrix} 0 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$4x_1 + 5x_2 = 0$$

$$4x_1 = -5x_2$$

$$\text{let } x_1 = t$$

$$x_2 = -\frac{4}{5}t$$

$$x = \begin{bmatrix} t \\ -\frac{4}{5}t \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{4}{5} \end{bmatrix} t$$

Simple soln

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} \quad \det(A - \lambda I) = \begin{bmatrix} 1 - \lambda & 0 \\ 2 & 4 - \lambda \end{bmatrix} = 0$$

$$(\lambda = 1, \lambda = 4) \text{ [eigen value of } A]$$

Eigen Value of $4A^{-1} + 3A + 2I$

$$= 4\lambda^{-1} + 3\lambda + 2$$

$$= \frac{4}{\lambda} + 3\lambda + 2$$

$$\lambda = 1 \Rightarrow (4 + 3) + (2) = 9$$

$$\lambda_{\text{new},1} = 9$$

$$\lambda = 4$$

$$(1 + 12 + 2) = 15 = \lambda_{\text{new},2}$$

Now use $\lambda_{n,1}$ and $\lambda_{n,2}$ to get Eigen Vectors



Ques) If $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$ find eigen values of $A^3 + 5A + 8I$

Eigen value of A :-

$$\lambda^3 - s_1 \lambda^2 + s_2 \lambda - |A|$$

$$\lambda^3 - 7\lambda + 6 = 0$$

$$\lambda^3 - 6\lambda - \lambda + 6 = 0$$

$$\lambda(\lambda^2 - 6) - (\lambda - 6) = 0$$

$$\lambda - 6 \quad \lambda = 1$$

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -7 & 6 \\ & & 1 & 1 & \\ \hline & 1 & 1 & -6 & \end{array}$$

$$(\lambda - 1)(\lambda^2 - \lambda - 6) = 0$$

$$(\lambda - 1)(\lambda - 3)(\lambda + 2)$$

for $\lambda = 1$ eigen value of $A^3 + 5A + 8I$
 $= 1 + 5 + 8 = 14$

for $\lambda = 3$ eigen value of $A^3 + 5A + 8I$

$$= 27 + 15 + 8 = 27 + 23 = 50$$

for $\lambda = -2$ eigen value of $A^3 + 5A + 8I$

$$= -8 + (-10) + 8 = -10$$

$$s_1 = 0$$

$$-1$$

$$s_2 = (-6) + (2) + (-3)$$

$$s_2 = -9 + 2$$

$$s_2 = -7$$

$$|A| = -6$$

$$6 \rightarrow 3 \times 2$$

$$-3\lambda + 2\lambda$$

$$(\lambda - 3)$$

$$0 = (\lambda - 1)(\lambda - 3)$$

$$+1 = \lambda \quad \lambda = 1$$

$$0 = \begin{bmatrix} 1 & x \\ 2 & y \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & + \end{bmatrix}$$

$$0 = 1 \times 2 + 1 \times 1$$

$$2 \times 2 = 1 \times 1$$

$$1 = 1 \times 1$$

$$1 = 1$$



• Statement of Cayley Hamilton Theorem

↳ Every square matrix satisfies its characteristic eqⁿ



for verifying, get the characteristic eqⁿ

$$\lambda^2 - s_1 \lambda + s_2 \lambda + |A|$$

s_1 = trace of A = sum of diagonal elements

$|A|$ = determinant

At you have to prove after putting A

in characteristic eqⁿ LHS = 0

for A^{-1} : multiply A^{-1} to the characteristic equation

• for A^t, A^S , multiply it to the characteristic eqⁿ !!!